MATH 142: Exam 3 **Name:** ______________________________
Spring —$_1$ 2026
04/16/2026
75 Minutes

Write your name on the appropriate line on the exam cover sheet. This exam contains 9 pages (including this cover page) and 6 questions. Check that you have every page of the exam. Answer the questions in the spaces provided on the question sheets. Be sure to answer every part of each question and show all your work and fully justify all your responses. If you run out of room for an answer, continue on the back of the page — being sure to indicate the problem number.

Question	Points	Score
1	20	
2	16	
3	16	
4	16	
5	16	
6	16	
Total:	100	

1. (20 points) Fill out the table of Maclaurin series below.

Function	Series	First Two Nonzero Terms	Interval of Convergence
$\dfrac{1}{1-x}$			
e^x			
$\sin(x)$			
$\cos(x)$			
$\ln(1+x)$			
$\arctan(x)$			

2. (16 points) Compute each of the following:

(a) $1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \cdots$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{(-\pi^2)^n}{(2n)!}$

(c) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{n\,3^n}$

(d) $4 - \dfrac{4}{3} + \dfrac{4}{5} - \dfrac{4}{7} + \dfrac{4}{9} - \dfrac{4}{11} + \cdots$

3. (16 points) Find the center, radius of convergence, and interval of converge for the following power series:

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n\,2^n}\,(x+5)^n$$

4. (16 points) Use a known Maclaurin series to find a power series representation for each of the expressions below. Be sure to fully simplify your series.

   (a) $2x\sin(x^3)$

   (b) $\dfrac{3}{2+x}$

(c) $5x^2e^{-x/5}$

(d) $\displaystyle\int \frac{\sin x}{x}\,dx$

5. (16 points) Showing all your work, complete the following parts:

(a) Find the first-degree Taylor polynomial, $T_1(x)$, for $\ln x$ centered at $x = 4$.

(b) Use the Taylor Remainder Theorem to find the maximum error in approximating $\ln 5$ by $T_1(5)$.

(c) Use the Taylor Remainder Theorem to find the maximum error in approximating $\ln x$ with $T_1(x)$ on $[2, 5]$.

6. (16 points) Find the Taylor $T(x)$ for $\dfrac{1}{(1-x)^2}$ centered at $x = 0$, and determine its interval of convergence. Show all your work. You may check your using a known Taylor series, but *no credit* will be given for only using the Taylor series of another function, i.e. a “memorized” Taylor series.